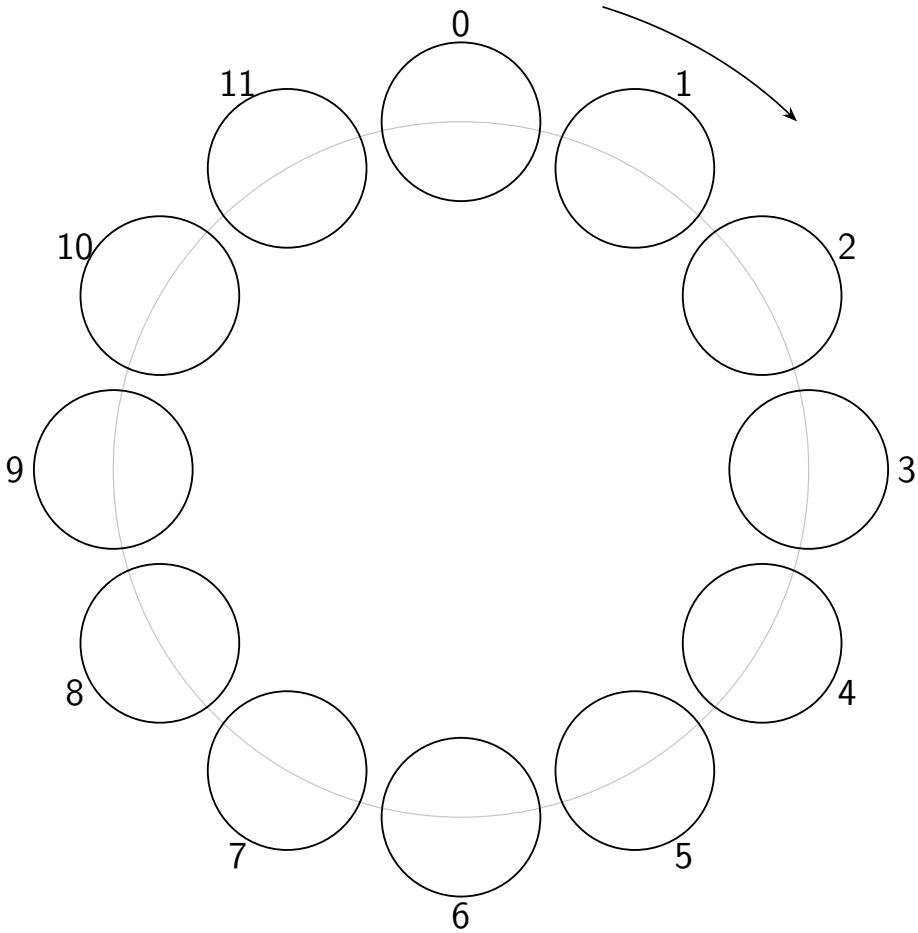
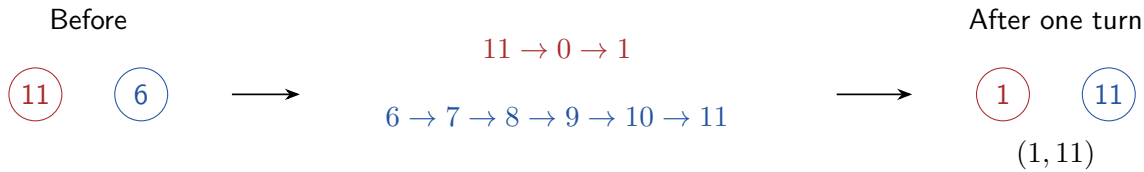
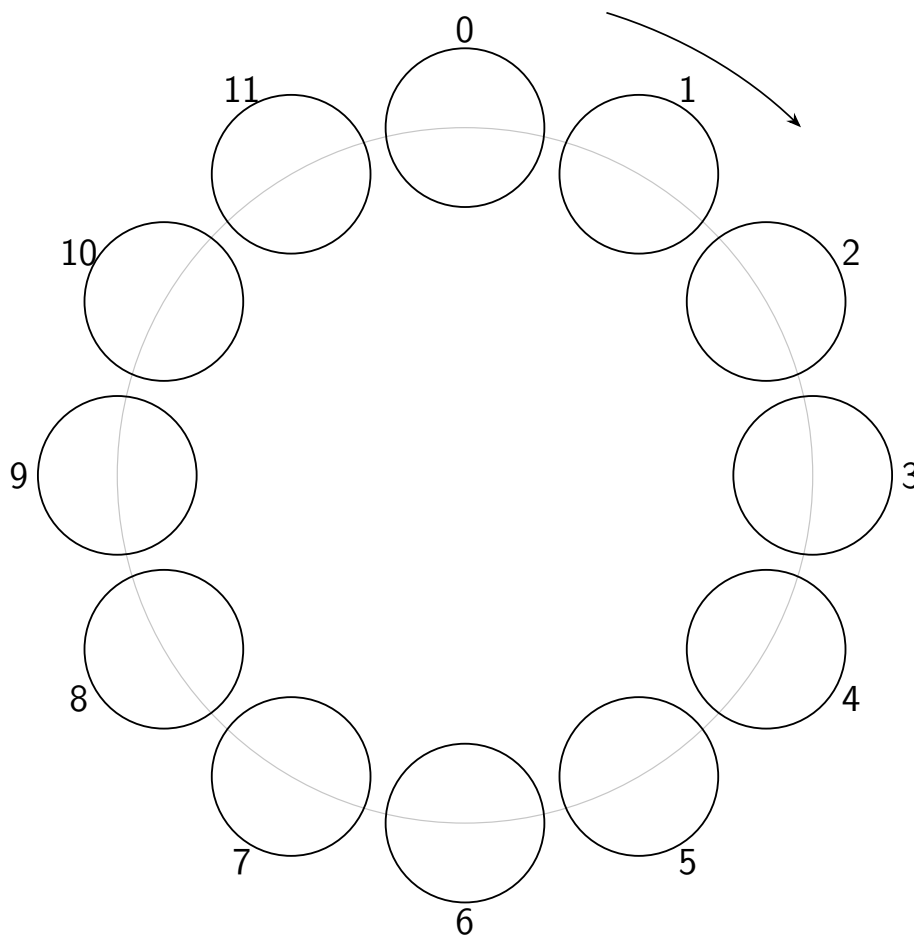
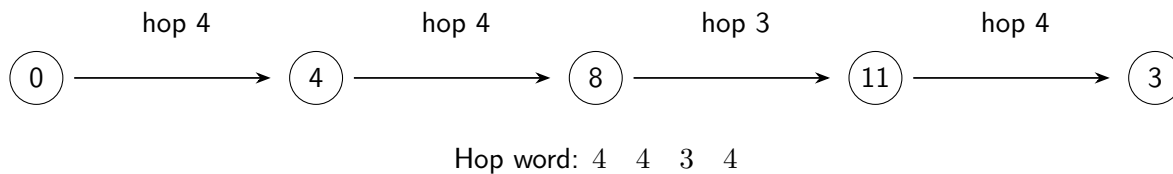


Problem 1: Each turn, red hops 2 circles clockwise and blue hops 5. Move both before you check. They meet only if they land on the same circle. Find the first meeting after at least one turn for each starting pair, if there is one. Then choose another starting pair. Record landings as (red, blue).



Red start	Blue start	Landings (red, blue)
0	0	
0	1	
0	3	

Problem 2: Start each route at 0. On each hop, choose either of the two allowed hop sizes and move clockwise. Find the circles you can reach with each pair of hop choices. Find shortest routes to 1 and to 2 whenever you can. Record a route by writing its hop sizes in order.



Hop choices	Circles you can reach	Hop word to 1	Hop word to 2
3 or 4			
4 or 6			

Problem 3: Place 3 red and 3 blue counters on the large ring. Two necklaces are the same if turning one ring while the other stays still makes them match. Do not flip a ring over. Draw every different necklace on the small rings. Which of your necklaces match themselves after a turn smaller than a full turn?

